Experiment No:	3
Subject:	Database Management Laboratory
Title:	Design and develop at least 5SQL queries for suitable database application using SQL DML statements: Insert and Select with operators and functions.
Roll No.:	
Class:	TE (E&TC)
Division:	
Signature:	

Group A- Database Programming Languages – SQL

Experiment No. -3

Title: Design and develop at least 5 SQL queries for suitable database application using SQL DML statements: Insert and Select with operators and functions.

Objective: To learn and understand DML statements in MySQL

Hardware requirements: Any CPU with Pentium Processor or similar, 256 MB RAM or more, 1 GB Hard Disk or more.

Software requirements: Windows Operating System, MySQL

Theory:

DML command

Data Manipulation Language (DML) statements are used for managing data in database. DML commands are not auto-committed. It means changes made by DML command are not permanent to database, it can be rolled back.

1) INSERT command

Insert command is used to insert data into a table. Following is its general syntax,

INSERT into table-name values(data1,data2,..)

Lets see an example,

Consider a table Student with following fields.

S_id S_Name age

INSERT into Student values(101,'Adam',15);

The above command will insert a record into Student table.

S_id S_Name age

101 Adam 15

2) UPDATE command

Update command is used to update a row of a table. Following is its general syntax,

UPDATE table-name set column-name = value where condition;

Lets see an example,

update Student set age=18 where s_id=102;

Example to Update multiple columns

UPDATE Student set s_name='Abhi',age=17 where s_id=103;

3) Delete command

Delete command is used to delete data from a table. Delete command can also be used with condition to delete a particular row. Following is its general syntax,

DELETE from table-name;

Example to Delete all Records from a Table

DELETE from Student;

The above command will delete all the records from Student table.

Example to Delete a particular Record from a Table

Consider Student table

DELETE from Student where s_id=103;

SQL Functions

SQL provides many built-in functions to perform operations on data. These functions are useful while performing mathematical calculations, string concatenations, sub-strings etc. SQL functions are divided into two catagories,

• Aggregrate Functions

• Scalar Functions

Aggregrate Functions

These functions return a single value after calculating from a group of values.Following are some frequently used Aggregrate functions.

1) AVG()

Average returns average value after calculating from values in a numeric column.

Its general Syntax is,

SELECT AVG(column_name) from table_name

e.g.

SELECT avg(salary) from Emp;

2) COUNT()

Count returns the number of rows present in the table either based on some condition or without condition.

Its general Syntax is,

SELECT COUNT(column_name) from table-name;

Example using COUNT()

Consider following Emp table

eid name age salary

401 Anu 22 9000

402 Shane 29 8000

SQL query to count employees, satisfying specified condition is,

SELECT COUNT(name) from Emp where salary = 8000;

3) FIRST()

First function returns first value of a selected column

Syntax for FIRST function is,

SELECT FIRST(column_name) from table-name

SQL query

SELECT FIRST(salary) from Emp;

4) LAST()

LAST return the return last value from selected column

Syntax of LAST function is,

SELECT LAST(column_name) from table-name

SQL query will be,

SELECT LAST(salary) from emp;

5) MAX()

MAX function returns maximum value from selected column of the table.

Syntax of MAX function is,

SELECT MAX(column_name) from table-name

SQL query to find Maximum salary is,

SELECT MAX(salary) from emp;

6) MIN()

MIN function returns minimum value from a selected column of the table.

Syntax for MIN function is,

SELECT MIN(column_name) from table-name

SQL query to find minimum salary is,

SELECT MIN(salary) from emp;

7) SUM()

SUM function returns total sum of a selected columns numeric values.

Syntax for SUM is,

SELECT SUM(column_name) from table-name

SQL query to find sum of salaries will be,

SELECT SUM(salary) from emp;

Scalar Functions

Scalar functions return a single value from an input value. Following are soe frequently used

Scalar Functions.

1) UCASE()

UCASE function is used to convert value of string column to Uppercase character.

Syntax of UCASE,

SELECT UCASE(column_name) from table-name

Example of UCASE()

SQL query for using UCASE is,

SELECT UCASE(name) from emp;

2) LCASE()

LCASE function is used to convert value of string column to Lowecase character.

Syntax for LCASE is:

SELECT LCASE(column_name) from table-name

3) MID()

MID function is used to extract substrings from column values of string type in a table.

Syntax for MID function is:

SELECT MID(column_name, start, length) from table-name

4) ROUND()

ROUND function is used to round a numeric field to number of nearest integer. It is used on Decimal point values. Syntax of Round function is,

SELECT ROUND(column_name, decimals) from table-name

Operators:

AND and OR operators are used with Where clause to make more precise conditions for fetching data from database by combining more than one condition together.

1) AND operator

AND operator is used to set multiple conditions with Where clause.

Example of AND

SELECT * from Emp WHERE salary < 10000 AND age > 25

2) OR operator

OR operator is also used to combine multiple conditions with Where clause. The only difference between AND and OR is their behaviour. When we use AND to combine two or more than two conditions, records satisfying all the condition will be in the result. But in case of OR, atleast one condition from the conditions specified must be satisfied by any record to be in the result.

Example of OR

SELECT * from Emp WHERE salary > 10000 OR age > 25

Set Operation in SQL

SQL supports few Set operations to be performed on table data. These are used to get meaningful results from data, under different special conditions.

3) Union

UNION is used to combine the results of two or more Select statements. However it will eliminate duplicate rows from its result set. In case of union, number of columns and datatype must be same in both the tables.

Example of UNION

select * from First

UNION

select * from second

4) Union All

This operation is similar to Union. But it also shows the duplicate rows.

Union All query will be like,

select * from First

UNION ALL

select * from second

5) Intersect

Intersect operation is used to combine two SELECT statements, but it only retuns the records which are common from both SELECT statements. In case of Intersect the number of columns and datatype must be same. MySQL does not support INTERSECT operator.

Intersect query will be,

select * from First

INTERSECT

select * from second

6) Minus

Minus operation combines result of two Select statements and return only those result which belongs to first set of result. MySQL does not support INTERSECT operator.

Minus query will be,

select * from First

MINUS

select * from second

Conclusion: